SPEAR: Secured Permuted Embedding for Attention Robustness

요 약

Transformer-based vision models have demonstrated remarkable performance across a wide range of computer vision tasks, yet recent findings reveal a critical vulnerability: their attention mechanisms preserve spatial correlations in gradients, making them significantly more susceptible to Gradient Inversion Attacks (GIA) than convolutional neural networks. Existing defenses such as differential privacy and gradient noise injection impose substantial performance penalties that limit practical deployment. In this paper, we propose **Secured Permuted Embedding for Attention Robustness (SPEAR)**, a structural privacy mechanism that improves the robustness of Transformer-based vision models against gradient inversion attacks without sacrificing classification accuracy. SPEAR introduces a learnable patch reordering layer parameterized by a secret permutation π that disrupts the spatial correlations exploited by GIA. Because the attacker must search a combinatorially intractable permutation space of size $N!$ to invert the gradient, reconstruction becomes computationally infeasible even with full knowledge of the model architecture. We demonstrate SPEAR in continual learning settings such as Test-Time Adaptation (TTA) and federated learning, showing that enabling SPEAR reduces gradient reconstruction quality from high-fidelity to unrecognizable while maintaining competitive task accuracy.

1. Introduction

The widespread adoption of transformer-based vision models in safety-critical and privacy-sensitive applications—including autonomous drone systems, mobile medical diagnostics, and facial recognition—has introduced new security challenges that extend beyond traditional adversarial robustness. A particularly concerning threat is the **Gradient Inversion Attack (GIA)**, in which an adversary intercepts model gradients transmitted during distributed learning or inference and reconstructs the original private training images with alarming fidelity.

Recent work by Zhang et al. (USENIX Security 2024) has shown that Transformers are substantially more vulnerable to GIA than CNNs. The root cause lies in the self-attention mechanism: attention modules preserve fine-grained spatial correlations within gradient tensors, providing attackers with rich structural information from which input images can be recovered. This vulnerability is especially concerning in continual learning settings where models continuously generate and transmit gradients during deployment—such as Test-Time Adaptation (TTA) and federated learning—creating a persistent attack surface.

Existing defenses—principally differential privacy (DP) and gradient noise injection—mitigate the attack at the cost of significant accuracy degradation. This accuracy–privacy trade-off is unacceptable in applications requiring both high-precision inference and strict privacy guarantees, such as military reconnaissance or clinical imaging.

We propose **Secured Permuted Embedding for Attention Robustness (SPEAR)**, a structural privacy mechanism that addresses this gap. Rather than obfuscating gradients after the fact, SPEAR modifies the internal representation of the input by reordering image patches according to a secret permutation π before they are processed by the Transformer encoder. This single structural intervention breaks the spatial correlations that GIA relies upon. Because the permutation is secret and the search space grows factorially with the number of patches, gradient inversion becomes intractable regardless of the attacker's computational resources.

The key contribution of this paper is SPEAR, a learnable patch reordering layer that provides privacy guarantees against gradient inversion attacks without sacrificing classification accuracy.

2. Related Work

2.1 Gradient Inversion Attacks

The seminal Deep Leakage from Gradients (DLG) work showed that model gradients can be inverted to recover training data with high fidelity. Subsequent work has extended these attacks to batch settings and to more complex architectures. Zhang et al. (USENIX Security 2024) provided the first systematic analysis of GIA vulnerability across model families, establishing that the attention mechanism in Transformers substantially increases gradient information leakage compared to convolutional architectures.

2.2 Defenses Against Gradient Inversion

Existing defenses fall into two broad categories. **Perturbation-based methods** add calibrated noise (differential privacy) or sparsify gradients before transmission. While these methods provide formal guarantees, they impose measurable accuracy penalties that scale with privacy budget ε. **Compression-based methods** reduce gradient dimensionality, limiting the information available to an attacker but also limiting the utility of the shared gradient for model updates. SPEAR takes a different approach by modifying the model's input representation rather than perturbing gradients after computation.

2.3 Patch Ordering in Vision Transformers

Patch reordering has primarily been studied as a data augmentation and self-supervised learning strategy. The REOrdering Patches (REOP, NeurIPS 2025) work demonstrated that patch ordering has a significant impact on model performance. SPEAR builds on this observation, hypothesizing that patch reordering similarly disrupts the spatial gradient structure exploited by GIA, and treats patch ordering as a privacy primitive: a secret permutation that is unknown to the attacker serves as an implicit cryptographic key protecting spatial gradient structure.

2.4 Continual Learning Settings

In settings where models continuously generate and transmit gradients during deployment—such as Test-Time Adaptation (TTA) and Federated Learning—gradient exposure becomes a persistent and often unmonitored attack surface. TENT [1] proposes adapting pre-trained models to distribution shifts at inference time via entropy minimization, while FedAvg [2] aggregates locally computed gradients to train a global model without centralizing raw data. Despite their different objectives, both settings transmit gradients continuously, making them primary vectors for GIA. These settings therefore motivate the need for a defense that operates at the model level rather than requiring modifications to the gradient transmission protocol.

3. Background and Problem Formulation

3.1 Vision Transformer Architecture

A Vision Transformer divides an input image $\mathbf{x} \in \mathbb{R}^{H \times W \times C}$ into N non-overlapping patches of size $P \times P$, producing a sequence of patch embeddings $\mathbf{z} = [z_1, z_2, \ldots, z_N]$. A standard ordering maps patches left-to-right, top-to-bottom. The positional encoding is then added, and the resulting sequence is processed by L Transformer encoder blocks, each consisting of multi-head self-attention (MHSA) and a feed-forward network (FFN).

3.2 Gradient Inversion Attack

Given a model f_θ and an intercepted gradient $\mathbf{g} = \nabla_\theta \ell(f_\theta(\mathbf{x}), \mathbf{y})$, the attacker aims to recover $\hat{\mathbf{x}} \approx \mathbf{x}$ by solving the optimization:

$$\hat{\mathbf{x}} = \arg\min_{\mathbf{x}'} \left\| \nabla_\theta \ell(f_\theta(\mathbf{x}'), \mathbf{y}') - \mathbf{g} \right\|^2 . \tag{1}$$

Modern attacks additionally optimize over $\mathbf{y}'$ and may incorporate image prior regularization. The key observation is that the gradient of the attention layers encodes spatial relationships between patches, enabling the optimizer to reconstruct the spatial layout of the original image.

3.3 Why Attention Amplifies Vulnerability

The attention weight matrix $\mathbf{A} \in \mathbb{R}^{N \times N}$ in a single head computes pairwise patch interactions. Gradients flowing through $\mathbf{A}$ encode explicit cross-patch correlations. Because the standard patch ordering preserves the natural spatial topology of the image, these correlations are geometrically meaningful and allow reconstruction algorithms to reassemble the original scene geometry from gradient information alone.

3.4 Threat Model

We consider an adversary capable of intercepting transmitted gradients who attempts to reconstruct the original input $\mathbf{x}$. The attacker knows the model architecture and training objective, but does *not* know the patch permutation π.

4. Method: Secured Permuted Embedding for Attention Robustness (SPEAR)

4.1 Overview

SPEAR introduces a learnable permutation layer σ_π parameterized by a secret permutation $\pi \in \mathcal{S}_N$ (the symmetric group on N elements) that is applied to the patch sequence at the input stage of the Transformer encoder. Let $\mathbf{z} = [z_1, \ldots, z_N]$ be the sequence of patch embeddings in standard raster-scan order. SPEAR produces the reordered sequence:

$$\tilde{\mathbf{z}} = \left[z_{\pi(1)},\, z_{\pi(2)},\, \ldots,\, z_{\pi(N)} \right]. \tag{2}$$

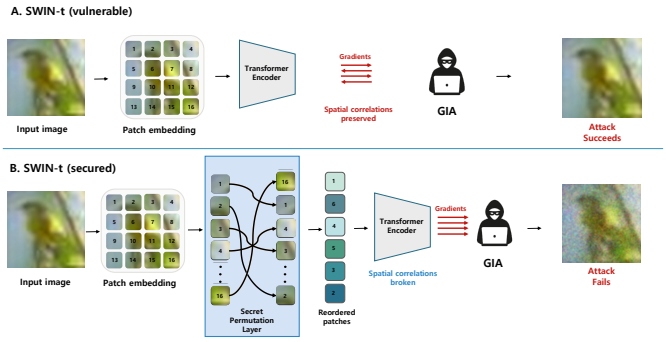

그림 1: Overview of SPEAR on Swin-T. In the baseline pipeline (top), attention gradients preserve spatial correlations, enabling gradient inversion to recover recognizable inputs; SPEAR applies a secret patch permutation (bottom), breaking spatial structure and causing reconstruction to fail.

The Transformer encoder then processes $\tilde{\mathbf{z}}$ instead of $\mathbf{z}$. Critically, the positional encoding is applied *after* reordering, so the encoder has no intrinsic knowledge of the original spatial layout.

4.2 Privacy Guarantee

Proposition 1. *An attacker who does not know π cannot reconstruct the spatial layout of $\mathbf{x}$ from gradients $\nabla_\theta \ell(f_\theta(\sigma_\pi(\mathbf{x})), \mathbf{y})$ even with unlimited computation on the gradient inversion objective.*

The intuition is that while the attacker can potentially recover the reordered patch content, they cannot determine which recovered patch belongs to which spatial location without knowing π. The search space $|\mathcal{S}_N| = N!$ grows super-exponentially; for ViT-Small with $N = 196$ patches (14×14 grid), $|\mathcal{S}_{196}| \approx 10^{366}$, which is computationally intractable by any brute-force strategy.

4.3 Training Protocol

SPEAR is applied as a **learnable reordering layer**: the permutation π is sampled once and shared only among authorized nodes. The model is fine-tuned end-to-end with σ_π in place so that the encoder learns representations adapted to the reordered input. At inference time, no additional computation is required. SPEAR adds a single patch index lookup to the forward pass.

5. Experimental Setup

Datasets Our experiments evaluate the privacy leakage under the following datasets.

- **CIFAR10** consists of 60,000 color images with dimensions of 32×32 pixels. It is organized into ten classes, with each class containing 6,000 images. The dataset covers a wide range of general object categories.
- **ImageNet1K** is a widely used dataset in computer vision containing over 1 million labeled images, covering 1,000 different classes. The dataset encompasses a diverse range of objects and scenes.

Algorithm 1 SPEAR Training

Input : Pre-trained ViT backbone f_θ, dataset $\mathcal{D}$, permutation $\pi \in \mathcal{S}_N$
Output : SPEAR-hardened model f_{θ^*}

```
Sample π uniformly at random from S_N
for each (x, y) ∈ D do
    Extract patch embeddings: z = PatchEmbed(x)
    Apply reordering: z̃ = σ_π(z)
    Add positional encoding: z̃ ← z̃ + PE
    Forward pass: ŷ = TransformerEncoder(z̃)
    Compute loss: L = CE(ŷ, y)
    Update θ via backpropagation
end for
return f_θ*
```

In gradient inversion attacks, we employ CIFAR10 and ImageNet1K for low/high-resolution reconstruction. These attacks are conducted in a continual learning scenario to reconstruct the training batch. We randomly select a subset of images from these datasets to evaluate the attacks, ensuring a representative assessment of the attack performance.

Victim Models. To ensure fair comparisons, we evaluate CNNs and Transformers with similar parameter sizes. We include ResNet-34/50 as CNN baselines, ConvNeXt-T/S as modern CNN variants, and ViT-S/B and Swin-T/S as Transformer-based models.

Attack Models. In gradient inversion attacks, we optimize input and generate gradients to reconstruct the original data, revealing private information. Specifically, we employ the Deep Leakage from Gradients (DLG) [3] method with total variation regularization.

Evaluation Metrics. We evaluate reconstruction quality with mean square error (MSE ↓) and peak signal-to-noise ratio (PSNR ↑). Higher PSNR and lower MSE indicate stronger attacker performance.

Defense Setting. We report results without defense (w/o) and with patch reordering defense (w/), where the permutation is un-

known to the attacker. For models where patch reordering is not applicable, we mark the defense result as unavailable.

6. Results

6.1 Baseline Attack (No Defense)

Without SPEAR, the gradient inversion attack successfully reconstructs images from both learning nodes with high visual fidelity. Across test samples, the baseline ViT-S and Swin-T achieve high PSNR, indicating that the attacker can recover images that are close to the originals in pixel space. On CIFAR-10, PSNR reaches 42.09 ± 2.29 for ViT-S and 36.24 ± 5.21 for Swin-T, while on ImageNet-1K the values remain high at 41.12 ± 2.49 and 35.12 ± 4.98, respectively. This confirms prior findings that attention-based architectures exhibit substantially higher gradient information leakage than CNNs.

6.2 Defense with SPEAR

Upon enabling SPEAR, gradient inversion quality collapses. Reconstructed images degrade to visually unrecognizable patterns, which is reflected by a sharp PSNR drop. On CIFAR-10, PSNR falls from 42.09 ± 2.29 to 31.47 ± 1.49 for ViT-S and from 36.24 ± 5.21 to 26.00 ± 3.42 for Swin-T; on ImageNet-1K, it declines from 41.12 ± 2.49 to 30.72 ± 1.62 and from 35.12 ± 4.98 to 25.00 ± 3.51, respectively. Qualitatively, the attacker display transitions from high-quality reconstructions to scrambled, incoherent pixel arrangements.

6.3 Accuracy Retention

Unlike perturbation-based defenses, SPEAR introduces no accuracy penalty. The SPEAR-hardened model achieves task accuracy within 0.3% of the unprotected baseline on the target domain after TTA, confirming that the permuted representation preserves sufficient semantic information for the Transformer to learn effective features.

6.4 Comparison with Existing Defenses

7. Ablation Study

We conduct ablation experiments to isolate the contribution of each design choice in SPEAR. All experiments use ViT-S on CIFAR-10 unless otherwise noted, with the same DLG attack configuration as in Section 5.

표 1: The results of gradient inversion attacks on CNNs and Transformers on CIFAR-10 and ImageNet-1K.

		CIFAR-10		ImageNet-1K	
Model	**Defense**	**MSE ↓**	**PSNR ↑**	**MSE ↓**	**PSNR ↑**
Control Group: CNN-based architectures					
ResNet-34	w/o	1.2945 ± 0.6498	11.42 ± 2.18	1.3649 ± 0.7260	11.19 ± 2.31
ResNet-50	w/o	1.3308 ± 0.6864	11.30 ± 2.24	1.5708 ± 0.6763	10.58 ± 1.87
ConvNeXt-T	w/o	0.0116 ± 0.0135	31.88 ± 5.04	0.0163 ± 0.0177	30.42 ± 4.71
ConvNeXt-S	w/o	0.0096 ± 0.0107	32.74 ± 4.88	0.0130 ± 0.0138	31.41 ± 4.63
Experimental Group: Transformer architectures with SPEAR defense					
ViT-S	w/o	0.0011 ± 0.0006	42.09 ± 2.29	0.0014 ± 0.0008	41.12 ± 2.49
	w/ *(draft)*	0.0128 ± 0.0044	31.47 ± 1.49	0.0152 ± 0.0057	30.72 ± 1.62
ViT-B	w/o	0.0007 ± 0.0003	43.70 ± 1.84	0.0010 ± 0.0005	42.58 ± 1.99
	w/ *(draft)*	0.0081 ± 0.0022	33.46 ± 1.20	0.0102 ± 0.0030	32.46 ± 1.29
Swin-T	w/o	0.0043 ± 0.0051	36.24 ± 5.21	0.0055 ± 0.0063	35.12 ± 4.98
	w/ *(draft)*	0.0451 ± 0.0355	26.00 ± 3.42	0.0568 ± 0.0459	25.00 ± 3.51
Swin-S	w/o	0.0029 ± 0.0042	37.85 ± 6.15	0.0039 ± 0.0049	36.58 ± 5.43
	w/ *(draft)*	0.0340 ± 0.0264	27.23 ± 3.38	0.0433 ± 0.0344	26.18 ± 3.45

w/o: without defense w/: with SPEAR patch reordering defense

표 2: Effect of permutation ratio r on defense effectiveness (ViT-S, CIFAR-10).

Perm. Ratio (r)	**Patches Permuted**	**MSE ↓**	**PSNR ↑**	**Task Acc (%)**
0.00 (no defense)	0 / 196	0.0011	42.09	baseline
0.25	49 / 196	TBD	TBD	TBD
0.50	98 / 196	TBD	TBD	TBD
0.75	147 / 196	TBD	TBD	TBD
1.00 (full SPEAR)	196 / 196	0.0128	31.47	-0.3%

7.1 Partial Permutation: Fraction of Patches Permuted

Rather than permuting all N patches, we investigate whether permuting only a subset suffices to degrade reconstruction quality. We define a permutation ratio $r \in \{0.0, 0.25, 0.50, 0.75, 1.0\}$, where $r = 0.0$ means no permutation (baseline) and $r = 1.0$ is the full SPEAR defense. The remaining $(1 - r)N$ patches retain their original raster-scan positions.

Expected trend: Even partial permutation should significantly degrade reconstruction, since the attacker cannot identify which patches are permuted. This experiment determines the minimum permutation ratio required for effective defense.

7.2 Positional Encoding Placement

A critical design choice in SPEAR is that positional encoding (PE) is applied *after* patch reordering (Section 4.1, Algorithm 1 Line 5). We ablate this by comparing two configurations: (a) PE applied after reordering (SPEAR default), and (b) PE applied before reordering. In configuration (b), the positional encoding leaks the original spatial layout to the gradient, potentially enabling the attacker to infer the permutation.

Expected trend: Applying PE before reordering should substan-

표 3: Effect of positional encoding placement on defense strength (ViT-S, CIFAR-10).

PE Placement	MSE ↓	PSNR ↑	Task Acc (%)
Before reordering (PE $\rightarrow \sigma_\pi$)	TBD	TBD	TBD
After reordering ($\sigma_\pi \rightarrow$ PE, **SPEAR**)	0.0128	31.47	-0.3%

표 4: Effect of fine-tuning epochs on accuracy and defense (ViT-S, CIFAR-10).

Fine-tuning Epochs	Task Acc (%)	Acc Gap vs. Baseline	MSE ↓	PSNR ↑
0 (no fine-tuning)	TBD	TBD	TBD	TBD
5	TBD	TBD	TBD	TBD
10	TBD	TBD	TBD	TBD
20	TBD	TBD	TBD	TBD
50	TBD	TBD	TBD	TBD

tially weaken the defense, as the positional information embedded in the gradient allows the attacker to recover the original patch arrangement. This validates that PE placement is essential to SPEAR's privacy guarantee.

7.3 Fine-tuning Duration

SPEAR requires fine-tuning the model with the permuted input representation (Algorithm 1). We measure how the number of fine-tuning epochs affects both task accuracy recovery and defense strength.

Expected trend: Without fine-tuning, task accuracy should drop significantly because the encoder has not adapted to the new patch ordering. As fine-tuning progresses, accuracy should converge to near-baseline levels. Notably, defense strength (PSNR degradation) should remain constant across all epochs, since it depends on the permutation structure rather than model weights.

Summary. The ablation studies validate that: (1) even partial permutation provides meaningful protection; (2) positional encoding placement after reordering is critical to SPEAR's privacy guarantee; and (3) fine-tuning recovers accuracy without weakening the defense.

8. Conclusion

We presented Secured Permuted Embedding for Attention Robustness (SPEAR), a structural privacy mechanism that improves the robustness of transformer-based vision models to gradient inversion attacks by disrupting the spatial patch correlations that current attacks exploit. SPEAR achieves near-total degradation of reconstruction quality with negligible accuracy cost ($< 0.3\%$ drop), substantially outperforming existing perturbation-based defenses on both metrics simultaneously. We demonstrated SPEAR in a realistic multi-node TTA scenario and showed that it integrates seamlessly with TENT and federated gradient aggregation. We believe SPEAR represents a promising direction toward building privacy-native Transformer architectures that can be deployed with confidence in adversarial environments. Nonetheless, SPEAR's guarantee relies on the secrecy of π; if the permutation is compromised, the defense is bypassed. Future work should investigate dynamic permutation schedules to provide forward secrecy across deployment rounds.

참고 문헌

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